

Lecture 1: Combinatorial Foundations of Probability

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Seminar 1

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Motivation

Probability starts with counting.

If all outcomes are equally likely,

$$P(A) = \frac{|A|}{|\Omega|}.$$

Therefore, many probability problems reduce to counting.

Example

A password consists of 4 digits.

How many passwords exist?

We have

$$10^4 = 10000.$$

The probability of guessing correctly in one attempt is

$$\frac{1}{10000}.$$